DMAIC quick reference

IES | Primer 3.0 | Synthetic worked example

Six Sigma DMAIC structures a process-improvement argument. AI can organise the report; evidence supports the conclusion.

Define and Measure

CTQ: fill volume in litres. Teaching limits: 9.4-10.6 L. Baseline samples 1-30; monitoring samples 31-35. Check units and missing values before analysis.

Analyse

Defect counts are 12, 8, 5 and 5. The first two categories account for 20/30=66.7 percent. Pareto prioritises investigation. Fishbone lists alternatives. Each 5-Whys link needs evidence.

Improve

Propose a bounded trial to investigate measurement and process conditions. State owner, expected effect, evidence and approval gate. A signal does not prove a root cause.

Control

Define the ongoing metric, review frequency, response to a signal and owner. Preserve baseline and source versions in the report.